Jeffrey Hoang

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github.com/jeffreyhoang | rag-portfolio-pi.vercel.app/

Education

San Jose State University Jan 2026 – Current
Master of Science in Artificial Intelligence GPA: 3.53/4.0

- **Relevant Coursework:** Artificial Intelligence and Data Engineering, Machine Learning

California State Polytechnic University, Pomona Aug 2022 – Dec 2025
Bachelor of Science in Computer Science GPA: 3.91/4.0

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Numerical Methods and Computing, Software Engineering, Design and Analysis of Algorithms, Data Structures and Advanced Programming, Database Systems
- **Awards:** Dean's List and President's List (all semesters)

Experience

Student Research Assistant Aug 2024 – Dec 2025
Computational Intelligence Lab | Department of Computer Science at Cal Poly Pomona
Supervisor: ***Dr. Hao Ji, Ph.D.****, Associate Professor*

- Developed web-based automation tools and REST APIs to control multi-camera GoPro setups using Python, OpenGoPro API, and USB/BLE/Wi-Fi protocols, increasing data collection efficiency
- Trained deep neural networks for 3D keypoint detection on the Human3.6M dataset using TensorFlow on NVIDIA A100 GPUs across high-performance computing (HPC) clusters, incorporating batch normalization, ReLU activation, dropout, and linear layers to improve accuracy and generalization
- Engineered a physics-based pipeline fitting Metrabs 3D keypoint predictions to biomechanical Locomujoco models, applying forward kinematics and optimization to estimate joint angles and body scaling parameters for anatomically consistent and physically accurate 3D human pose alignments

Projects

ASL Recognition | Python, PyTorch, NumPy, Flask, MediaPipe, OpenAI API Jan 2026 – May 2026

- Extracted 126-dimensional skeletal features per frame using MediaPipe 3D hand landmarks, then engineered frame-to-frame velocity features to produce a 252-dimensional input vector, improving model accuracy and generalization
- Trained a BiLSTM with temporal attention achieving 73% test accuracy and a Transformer achieving 69.9% test accuracy on a 56-class filtered WLASL dataset, both exceeding the post-only SOTA benchmark of 55-60%
- Integrated a sliding window inference pipeline with LLM post-processing to convert isolated sign predictions into grammatically coherent sentences in real time

RAG Portfolio | Python, FastAPI, React, LangChain, Pinecone, OpenAI API Jan 2026 – May 2026

- Deployed a production RAG assistant on my portfolio website, enabling recruiters to ask natural language questions answered with cited, grounded responses pulled from my personal resume and documents
- Designed an ingestion pipeline that loads, chunks, and embeds documents into 512-token chunks with 50-token overlap using OpenAI's text-embedding-3-small model, storing vectors in ChromaDB (dev) and Pinecone (prod)
- Engineered a retrieval pipeline with GPT-4o-mini query rewriting, top-10 vector search via ChromaDB/Pinecone, and ms-marco-MiniLM-L-6-v2 cross-encoder reranking to return the 3 most relevant chunks per query, with multi-turn memory maintained via a sliding window of the last 10 conversation turns

Multimodal Sentiment Analysis | Python, PyTorch, Torchvision, Pandas, Scikit-learn Jan 2026 – May 2026

- Fine-tuned BERT (full fine-tune) for text and ResNet-18 (last 2 layers + fully connected layer unfrozen) for image feature extraction on the MVSA-Single dataset of social media post pairs labeled positive, neutral, or negative
- Evaluated four multimodal fusion strategies, including score-level, embedding-level, scalar-gated, and vector-gated, against unimodal text-only and image-only baselines across accuracy, precision, recall, and F1-score
- Achieved 79.43% test accuracy with score-level and vector-gated fusion models, outperforming the text-only baseline (77.38%) and image-only baseline (65.04%) and demonstrating the benefit of multimodal fusion for sentiment classification.

Skills

Languages: Python, JavaScript, TypeScript, SQL **ML/AI:** PyTorch, TensorFlow/Keras, MediaPipe, LangChain, Claude Code, HuggingFace, OpenAI API, NumPy, Pandas, Scikit-learn **Full-Stack:** FastAPI, Flask, Next.js, React, Tailwind CSS, Git